

6.4.3

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Question

Find the shortest distance between the lines:

$$\begin{aligned}\mathbf{r} &= 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k}) \\ \mathbf{r} &= 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})\end{aligned}$$

Theoretical Solution

Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be the points on the given lines respectively.

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} + k_1 \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \text{ and } \mathbf{x}_2 = \begin{pmatrix} 7 \\ 0 \\ -6 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$\text{Let } \mathbf{A} = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 7 \\ 0 \\ -6 \end{pmatrix}$$

$$\text{Let } \mathbf{M} = \begin{pmatrix} 3 & 1 \\ 2 & 2 \\ 6 & 2 \end{pmatrix}$$

$$(\mathbf{M} \mathbf{B} - \mathbf{A}) = \begin{pmatrix} 3 & 1 & 5 \\ 2 & 2 & 5 \\ 6 & 2 & -7 \end{pmatrix} \quad (1)$$

Theoretical Solution

Row Transformation-1: $R_3 \rightarrow R_3 - 2R_1$

$$\begin{pmatrix} 3 & 1 & 5 \\ 2 & 2 & 5 \\ 0 & 0 & -17 \end{pmatrix} \quad (2)$$

Row Transformation-2: $R_2 \rightarrow R_2 - \frac{2}{3}R_1$

$$\begin{pmatrix} 3 & 1 & 5 \\ 0 & 4/3 & 5/3 \\ 0 & 0 & -17 \end{pmatrix} \quad (3)$$

Therefore, The Rank is 3 $\Rightarrow$ The Lines are Skew Lines.

$$\text{Let } \mathbf{K} = \begin{pmatrix} k_1 \\ -k_2 \end{pmatrix} \quad (4)$$

Theoretical Solution

$$(\mathbf{M}^T \mathbf{M})\mathbf{K} = \mathbf{M}^T(\mathbf{B} - \mathbf{A}) \quad (5)$$

$$\begin{pmatrix} 49 & 19 \\ 19 & 9 \end{pmatrix} \mathbf{K} = \begin{pmatrix} -17 \\ 1 \end{pmatrix} \quad (6)$$

The Augmented Matrix from Equation 0.6,

$$\left(\begin{array}{cc|c} 49 & 19 & -17 \\ 19 & 9 & 1 \end{array} \right) \quad (7)$$

After Row Reductions,

$$\left(\begin{array}{cc|c} 1 & 0 & -43/20 \\ 0 & 1 & 93/20 \end{array} \right) \quad (8)$$

$$\therefore \mathbf{K} = \begin{pmatrix} -43/20 \\ 93/20 \end{pmatrix} \quad (9)$$

Theoretical Solution

$$\therefore k_1 = -43/20 \text{ and } k_2 = -93/20 \quad (10)$$

From Equation 0.10,

$$\mathbf{x}_1 = \begin{pmatrix} -89/20 \\ -93/10 \\ 119/10 \end{pmatrix} \text{ and } \mathbf{x}_2 = \begin{pmatrix} -47/20 \\ -93/10 \\ -153/10 \end{pmatrix} \quad (11)$$

The Minimum Distance between the given skew lines is $\|\mathbf{x}_2 - \mathbf{x}_1\|$

$$\|\mathbf{x}_2 - \mathbf{x}_1\| = \sqrt{(\mathbf{x}_2 - \mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1)} = \frac{17}{\sqrt{5}} \quad (12)$$

$$\text{The Minimum Distance between the given Lines} = \frac{17}{\sqrt{5}} \text{ units} \quad (13)$$

Skew Lines

